

Lesson 5 Ros Basic Case 2- Write a Simple Publisher and Subscriber

1. Write the publisher node

In this lesson, we are going to create a talker node and the node source code is making references to github open source code.

- 1) Enter “**roscd beginner_hiwonder**” command and press “Enter”. Then enter beginner_hiwonder software package newly created.

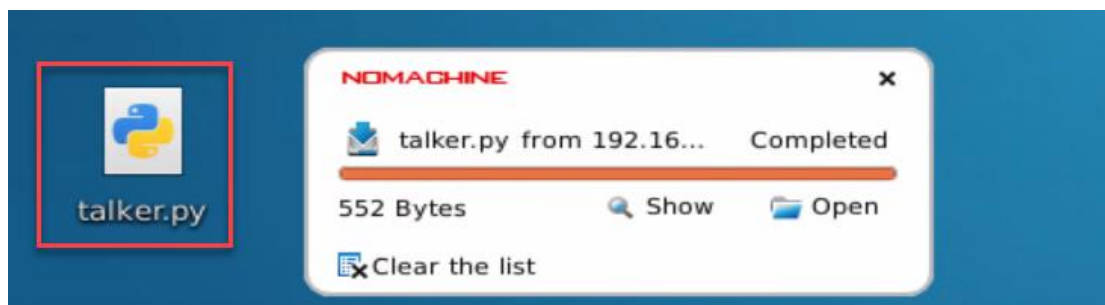
```
ubuntu@ubuntu:~/Desktop$ roscd beginner_hiwonder
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder$
```

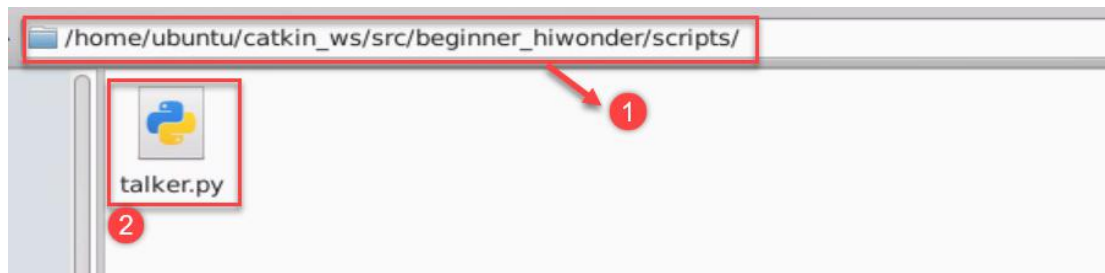
- 2) Enter “**mkdir scripts**” command and press “Enter”. Then create a scripts directory to save Python script.

```
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder$ mkdir scripts
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder$
```

- 3) Drag the code file of publisher node, talker.py, to the desktop of NoMachine. The code file is under ArmPi FPV data path “**4. Advanced Program Lessons \ ROS Basic Lesson \Ros Related Program**”

Then, move the talker.py to "/home/ubuntu/catkin_ws/src/beginner_hiwonder/scripts/".





- 4) On the terminal that has just opened, enter “**cd scripts**” command and press “Enter” to enter a new scripts directory.

```
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder$ cd scripts
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder/scripts$
```

- 5) Enter “**chmod +x talker.py**” command and press “Enter”. Give execute permission to talker.py.

```
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder/scripts$ chmod +x talker.py
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder/scripts$
```

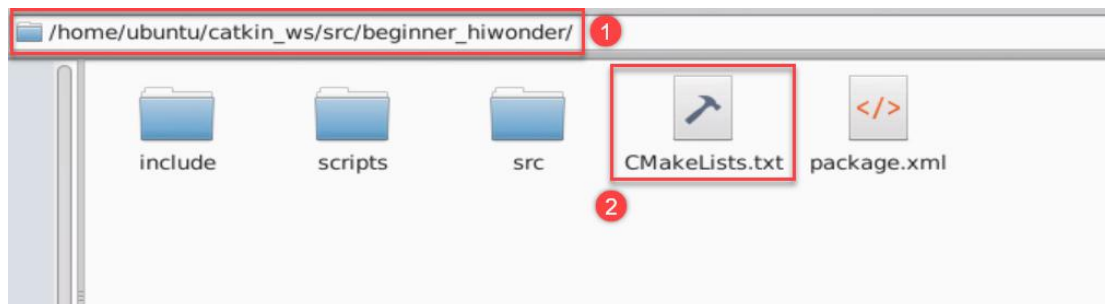
- 6) If wanting to revise or edit talker.py file, you need to enter “**roscd beginner_hiwonder talker.py**” command and press “Enter”. Then press “i”. When finishing revising, press “Esc” and enter “: wq” and save to quit.

```
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder/scripts$ roscd beginner_hiwonder talker.py
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder/scripts$
```

```
1 #!/usr/bin/env python
2
3 import rospy
4 from std_msgs.msg import String
5
6 def talker():
7     pub = rospy.Publisher('chatter', String, queue_size=10)
8     rospy.init_node('talker', anonymous=True)
9     rate = rospy.Rate(10) # 10hz
10    while not rospy.is_shutdown():
11        hello_str = "hello world %s" % rospy.get_time()
12        rospy.loginfo(hello_str)
13        pub.publish(hello_str)
14        rate.sleep()
15
16 if __name__ == '__main__':
17     try:
18         talker()
19     except rospy.ROSInterruptException:
20         pass
21
```

- 7) Add the following content to the CMakeLists.txt file. Ensure that Python scripts are installed correctly and that the suitable Python interpreter is used.

```
catkin_install_python(PROGRAMS scripts/talker.py DESTINATION
${CATKIN_PACKAGE_BIN_DESTINATION})
```



```
## Add gtest based cpp test target and link libraries
# catkin_add_gtest(${PROJECT_NAME}-test test/test_beginner_hiwonder.cpp)
# if(TARGET ${PROJECT_NAME}-test)
#   target_link_libraries(${PROJECT_NAME}-test ${PROJECT_NAME})
# endif()

## Add folders to be run by python nosetests
# catkin_add_nosetests(test)

catkin_install_python(PROGRAMS scripts/talker.py
  DESTINATION ${CATKIN_PACKAGE_BIN_DESTINATION}
)
```

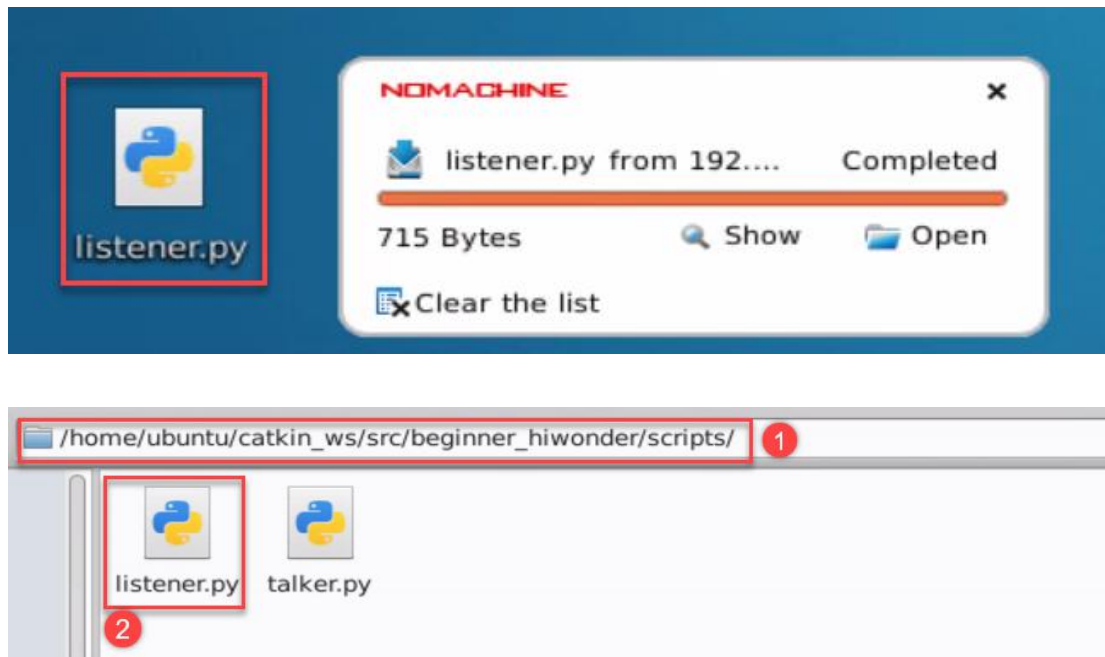
2. Write subscriber node

In this lesson, we are going to create a listener node and the node source code is making references to github open source code.

- 1) Drag the code file of listener node, listener.py, to the desktop of NoMachine. The listener node is under ArmPi FPV data path “4. Advanced Program Lessons \ ROS Basic Lesson \Ros Related Program”

Then, move listener.py to

“/home/ubuntu/catkin_ws/src/beginner_hiwonder/scripts/”.



- 2) Open command line terminal to enter “**cd ~/catkin_ws/src/beginner_hiwonder/scripts/**” command and press “Enter” to enter scripts directory newly created.

```
ubuntu@ubuntu:~$ cd ~/catkin_ws/src/beginner_hiwonder/scripts/  
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder/scripts$
```

- 3) Enter “**chmod +x listener.py**” command and press “Enter”. Give execute permission to listener.py.

```
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder/scripts$ chmod +x listener.py  
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder/scripts$
```

- 4) If wanting to revise or edit listener.py file, you need to enter “**rosed beginner_hiwonder listener.py**” command and press “Enter”. Then press “i”. When finishing revising, press “Esc” and enter “: **wq**” and save to quit.

```
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder/scripts$ rosed beginner_hiwonder listener.py  
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder/scripts$
```

```

1 #!/usr/bin/env python
2 import rospy
3 from std_msgs.msg import String
4
5 def callback(data):
6     rospy.loginfo(rospy.get_caller_id() + "I heard %s", data.data)
7
8 def listener():
9
10    # In ROS, nodes are uniquely named. If two nodes with the same
11    # name are launched, the previous one is kicked off. The
12    # anonymous=True flag means that rospy will choose a unique
13    # name for our 'listener' node so that multiple listeners can
14    # run simultaneously.
15    rospy.init_node('listener', anonymous=True)
16
17    rospy.Subscriber("chatter", String, callback)
18
19    # spin() simply keeps python from exiting until this node is stopped
20    rospy.spin()
21
22 if __name__ == '__main__':
23     listener()

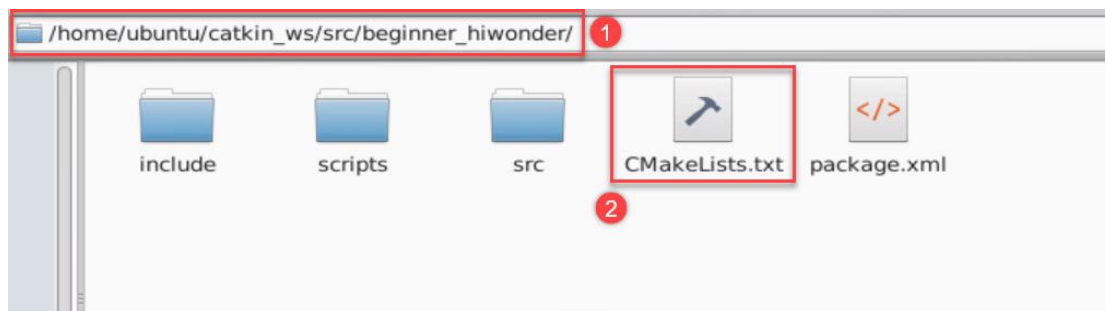
```

- 5) Add the following content to the CMakeLists.txt file. Ensure that Python scripts are installed correctly and that the suitable Python interpreter is used.

```

catkin_install_python(PROGRAMS scripts/talker.py scripts/listener.py
DESTINATION ${CATKIN_PACKAGE_BIN_DESTINATION})

```



```

## Add folders to be run by python nosetests
# catkin_add_nosetests(test)

catkin_install_python(PROGRAMS scripts/talker.py
DESTINATION ${CATKIN_PACKAGE_BIN_DESTINATION}
)

catkin_install_python(PROGRAMS scripts/talker.py scripts/listener.py DESTINATION $
{CATKIN_PACKAGE_BIN_DESTINATION}
)

```


3. Verify publisher and subscriber

- 1) To ensure the Python code is generated automatically for created message and service, we need to return to catkin workspace and enter “cd ~/catkin_ws” command, then press “Enter”. Next, enter “catkin_make” command and press “Enter”.

```
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder/scripts$ cd ~/catkin_ws
ubuntu@ubuntu:~/catkin_ws$

ubuntu@ubuntu:~/catkin_ws$ catkin_make
Base path: /home/ubuntu/catkin_ws
Source space: /home/ubuntu/catkin_ws/src
Build space: /home/ubuntu/catkin_ws/build
Devel space: /home/ubuntu/catkin_ws/devel
Install space: /home/ubuntu/catkin_ws/install
```

- 2) Enter “roscore” command and start the node manager. If it is opened, the following prompt will appear.

```
ubuntu@ubuntu:~/catkin_ws$ roscore
... logging to /home/ubuntu/.ros/log/b5a8bc06-9034-11eb-bbc3-f34c134976d1/roslaunch-ubuntu-4256.log
Checking log directory for disk usage. This may take a while.
Press Ctrl-C to interrupt
Done checking log file disk usage. Usage is <1GB.

started roslaunch server http://localhost:35053/
ros_comm version 1.14.10

SUMMARY
=====
PARAMETERS
* /rostdistro: melodic
* /rosversion: 1.14.10

NODES
RLErrorException: roscore cannot run as another roscore/master is already running.
Please kill other roscore/master processes before relaunching.
The ROS_MASTER_URI is http://localhost:11311/
The traceback for the exception was written to the log file
```

- 3) Before running the program, please make sure the setup.*sh file of the workspace is sourced after calling catkin_make. If not, you need to return to the created workspace and make setup.*sh take effect.

```
ubuntu@ubuntu:~/catkin_ws$ source ./devel/setup.bash
ubuntu@ubuntu:~/catkin_ws$
```

- 4) Enter “roslaunch beginner_hiwonder talker.py” command to run publisher of talker. And you can quit the publisher node by pressing “Ctrl+C”.

```
ubuntu@ubuntu:~/catkin_ws$ rosrun beginner hiwonder talker.py
[INFO] [1616985959.703996]: hello world 1616985959.7
[INFO] [1616985959.803931]: hello world 1616985959.8
[INFO] [1616985959.904136]: hello world 1616985959.9
[INFO] [1616985960.003897]: hello world 1616985960.0
[INFO] [1616985960.103892]: hello world 1616985960.1
[INFO] [1616985960.203944]: hello world 1616985960.2
[INFO] [1616985960.304175]: hello world 1616985960.3
[INFO] [1616985960.404285]: hello world 1616985960.4
[INFO] [1616985960.503878]: hello world 1616985960.5
^Cubuntu@ubuntu:~/catkin_ws$
```

- 5) Open a new command line terminal and enter “roslaunch beginner_hiwonder listener.py” to run subscriber of listener. And you can quit the subscriber node by pressing “Ctrl+C”.

```
ubuntu@ubuntu:~/catkin_ws$ roslaunch beginner_hiwonder listener.py
[INFO] [1616985959.710734]: /listener_18169_1616985952432I heard hello world 1616985959.7
[INFO] [1616985959.811716]: /listener_18169_1616985952432I heard hello world 1616985959.8
[INFO] [1616985959.918876]: /listener_18169_1616985952432I heard hello world 1616985959.9
[INFO] [1616985960.011054]: /listener_18169_1616985952432I heard hello world 1616985960.0
[INFO] [1616985960.110876]: /listener_18169_1616985952432I heard hello world 1616985960.1
[INFO] [1616985960.211918]: /listener_18169_1616985952432I heard hello world 1616985960.2
[INFO] [1616985960.311063]: /listener_18169_1616985952432I heard hello world 1616985960.3
[INFO] [1616985960.410184]: /listener_18169_1616985952432I heard hello world 1616985960.4
[INFO] [1616985960.511627]: /listener_18169_1616985952432I heard hello world 1616985960.5
^Cubuntu@ubuntu:~/catkin_ws$
```

Note: 1. The subscriber node cannot subscribe message until starting the publisher node.

2. If you need to receive all publisher message, you can start the subscriber node first, then turn on publisher node.